

Tafila Technical University
Faculty of Information Technology and Communication
Department of Computer Science

Course Information

Course Title: AI Programming

Course Code: 0602221

Homework Number: HW 2

Title: Smart Access Control System

Due Date: (To be announced by instructor)

Objectives

- Apply conditional statements (*if / elif / else*)
- Use logical operators: *and, or, not*
- Understand *Boolean expressions*
- Perform *basic input validation*
- Work with different data types
- Analyze how Python evaluates expressions

Assignment Description

You are required to implement a Smart Access Control System for a secure research lab. The system determines whether a user is allowed access based on several conditions such as age, access level, and membership status.

Requirements

Create a Python program named: access_control_YourId.py

Program Input

1. Age (integer)
2. Access Level (1-5)
3. Security Code (integer)

4. Membership Status (True/False)

Input Validation

Age ≥ 18

Access Level between 1–5

Security Code > 0

If invalid: "*Invalid input detected. Access denied.*"

Then terminate the program immediately.

Access Rules

Rule 1: Access Level ≥ 4 AND Membership is **True**

Rule 2: Access Level ≥ 3 AND Security Code is **even**

Rule 3: Deny if Age < 21 AND Access Level < 5

Final Decision

Print "Access Granted" or "Access Denied"

Sample Run:

```
Enter age: 25
Enter access level (1-5): 3
Enter security code: 2468
Enter membership status (True/False): False

Access Granted
```

```
Enter age: 19
Enter access level (1-5): 2
Enter security code: 1357
Enter membership status (True/False): True

Access Denied
```

Constraints:

Do NOT use: **loops (for, while), lists or try/except**. Use only **if, elif, else** and logical operators (**and, or, not**)

Conceptual Questions

1. Explain short-circuit evaluation.

Explain what happens when evaluating a logical expression using and if the first condition is False.

2. *Difference between and / or.*

3. Truthy vs Falsy values.

What are truthy and falsy values in Python?

Give at least three examples of each.

4. What will be the result of the following expression, and why?

True and 6.2

5. *Why logical expressions return non-Boolean values.*

6. *Importance of input validation.*